

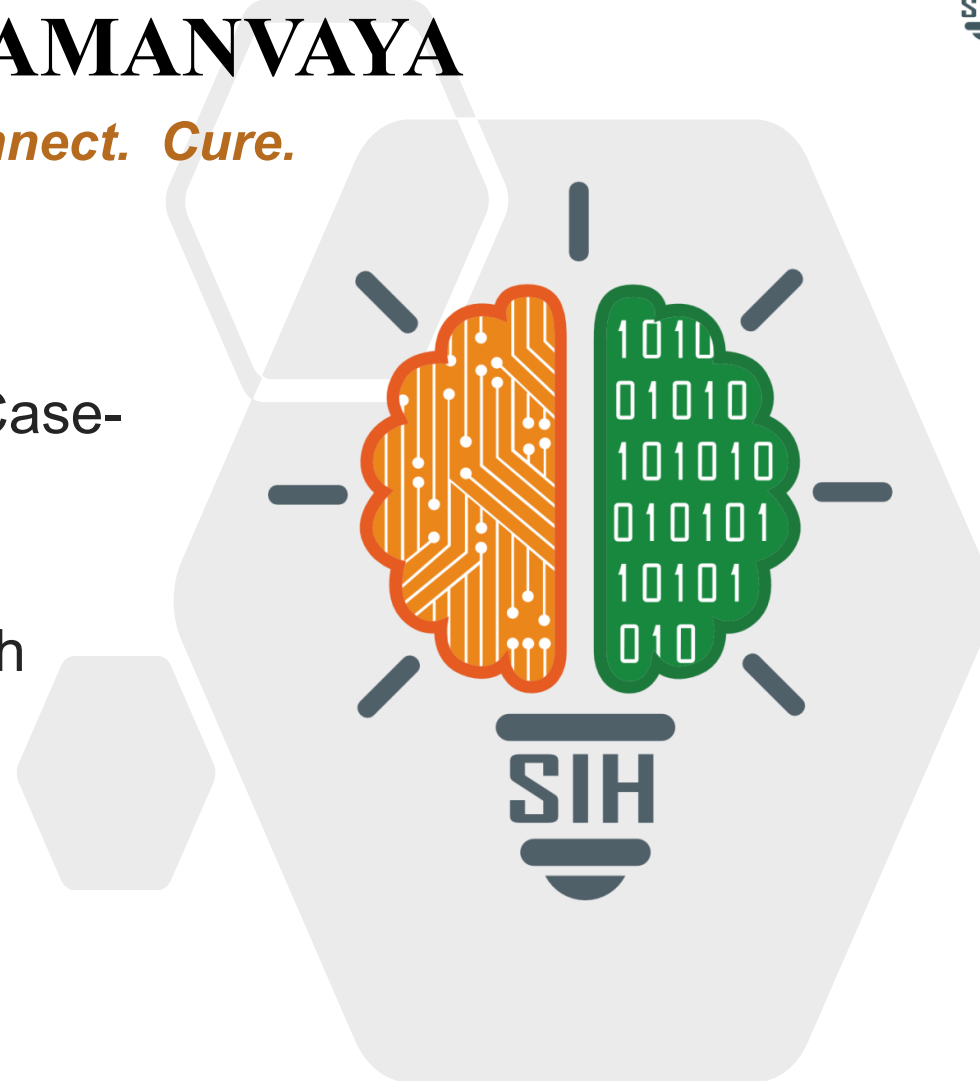
SMART INDIA HACKATHON 2026



PROJECT SAMANVAYA

Collect. Connect. Cure.

- **Problem Statement ID** – SIH26047
- **Problem Statement Title**- Patient Case-Taking Software
- **Theme**- MedTech /BioTech / HealthTech
- **PS Category**- Software
- **Team ID**- 125797
- **Team Name** : Team Panchajanyam



PROJECT SAMANVAYA

Collect. Connect. Cure.

Proposed Solution :

- ❖ A 3-portal software platform (Patient Portal, OPD Kiosk & Doctor-Staff Suite) that lets any patient build a complete medical history through voice and touch in 22 – Indian local languages, digitize their existing documents, and hand the doctor a ready, structured case sheet — before the consultation even begins.
- ❖ Features an Autonomous Floating NLP AI Assistant that provides end-to-end application guidance, evaluating eligible govt schemes, calculating 80%+ Jan Aushadhi generic savings and enforcing strict patient data privacy and safety via Govt ABDM Sandbox Integration

WHAT IT DOES



Voice + Touch History Intake

Guided Symptom questioning in 22 Languages, plus a dedicated AYUSH Dashavidha Pariksha mode for Ayurvedic OPDs.



Document Digitization

Reads handwritten prescriptions through Multi OCR, lab reports and discharge summaries, then auto-dates them into one chronological timeline and links it with ABHA ID.



Structured Case Summary

Builds a physician-ready Chief Complaint → HPI → History → ROS draft the doctor edits and confirms in seconds.



Govt ABHA & Privacy via ABDM

Secures patient data under strict privacy laws, creates and links to ABHA IDs, and syncs directly with hospital software

WHERE SAMANVAYA GOES FURTHER



Autonomous Floating Assistant

Auto-fills medical forms free hand, switches app portals via voice, and provides end-to-end guidance throughout the platform



Govt Schemes & JAN Aushadhi Savings

Guides patients and families step-by-step to claim eligible Govt Health (both Central & State Govt) Schemes, promotes generic medicine with estimated costs & locates nearest Jan Aushadhi kendras,



Unified 3-Portal Ecosystem

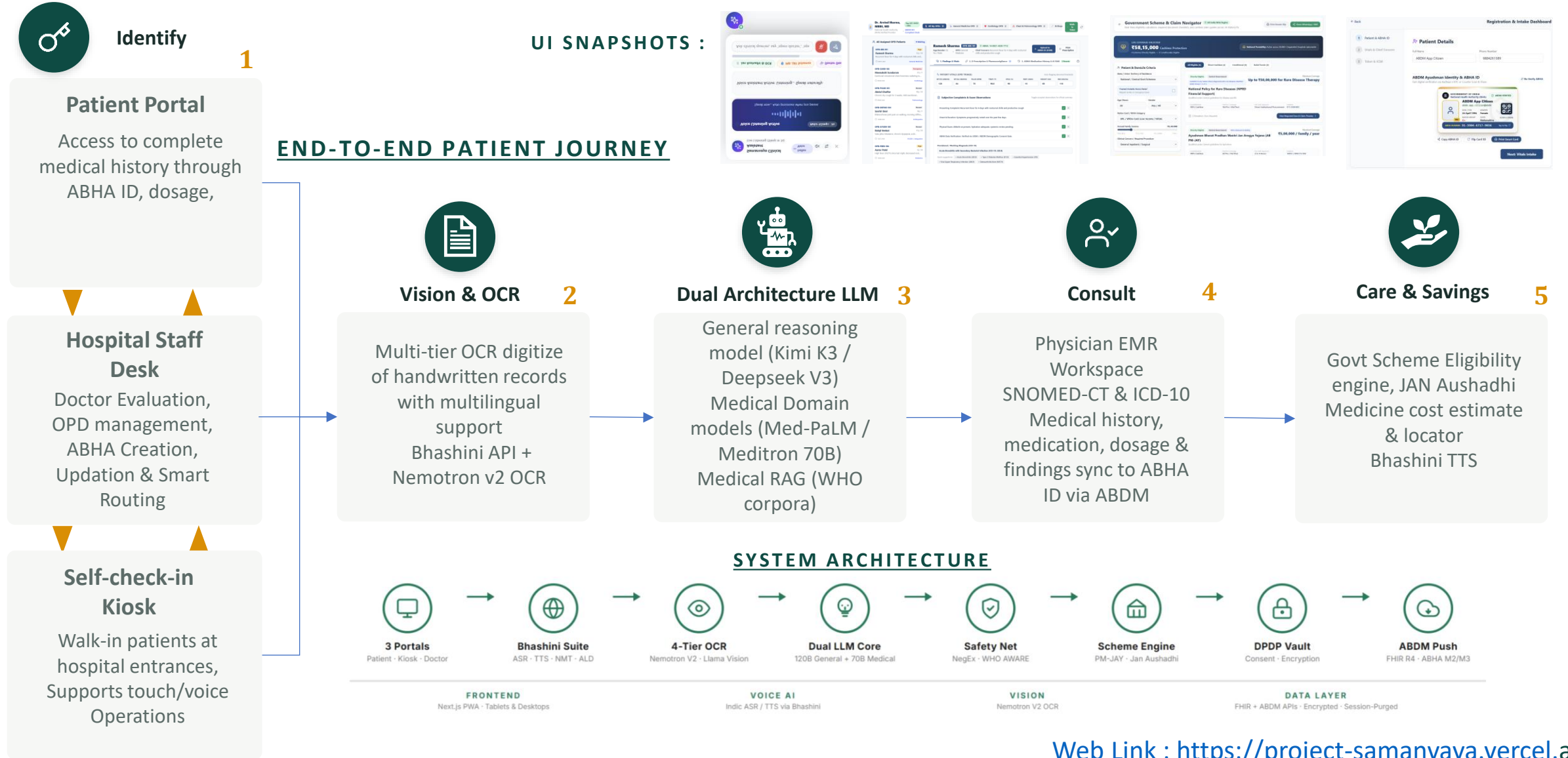
Seamlessly connects Patient portal, OPD Kiosk & Doctor-Staff suite so patient data flows instantly into Physician workflows



Clinical Safety, Privacy Edge AI Net

Integrates Herb-Drug polypharmacy checks, WHO AWARE antibiotic audits, Vernacular audio dosage guidance, offline Edge syncing during internet blackouts.

TECHNICAL APPROACH





Technical Feasibility

- Built on proven building blocks — conversational LLMs, Autonomous AI, OCR, and India's own Bhashini voice APIs.
- ABDM's FHIR APIs and ABHA already provide the interoperability layer, Samanvaya plugs into it.
- Runs on the low-cost tablets/desktops already present in OPDs — no new hardware.



Economic Viability

- Software-only deployment — one platform per OPD desk, not per-patient hardware.
- Directly slashes out of pocket expenditure (OOPE) for citizens by preventing repeated lab tests, files and expensive brand prescriptions
- Calculate 80% Generic drug savings and maps nearest Kendras



Operational Viability

- Sits ahead of the consultation, not inside it — intake happens in the waiting room.
- Doctor stays the final authority — every AI summary is a draft to accept, edit or reject.
- Offline-first design keeps it usable where hospital connectivity is unreliable.

POTENTIAL CHALLENGES & HOW WE ADDRESS THEM

Challenge	Mitigation Strategy
Low digital literacy / elderly patients	Autonomous floating NLP assistant, Dual voice + touch input, audio-guided consent, zero-training icon-based UI.
Keeping medical & scheme data current	Scheduled sync with ICMR / PMBJP / scheme sources — versioned, not hardcoded.
OCR accuracy on poor handwriting	Human-in-the-loop review — doctor edits and confirms before anything saves.
Data privacy & consent at scale	DPDP Act-compliant flow; raw scans purged immediately after structuring.

IMPACT AND BENEFITS



For Patients

- History captured once — no repeating the same story at every visit.
- Care in their own language, usable by low-literacy and elderly patients via their local voice support
- Clear visibility into scheme eligibility and lower-cost generic alternatives.



For Doctors

- Walks in with the reading already done — not the recording of it, helps in exposing hidden history or symptoms.
- AI-flagged red flags and drug interactions add a second layer of safety.
- Summaries are ABDM/FHIR-ready — no extra transcription step.



For the Health System

- Digitizes the ABDM “first mile” that currently has no software solution.
- AYUSH-aware from day one — one platform for allopathic and Ayurvedic OPDs.
- Reclaims clinical time without adding headcount.



80%+ Reduction in out-of-pocket expenses



Elimination of Lost Files & Duplicate Tests



5x Increase in Hospital OPD Capacity



Universal Health Care Equity

SUPPORTING RESEARCH



Irving G, et al. “International variations in primary care physician consultation time: a systematic review of 67 countries.”

BMJ Open, 2017;7(10):e017902 • doi.org/10.1136/bmjopen-2017-017902



Problem Statement SIH26047 — “Patient Case-Taking Software”

All India Institute of Ayurveda, Ministry of Ayush • Smart India Hackathon 2026



Medical Patient Appointments management

IEEE Conference Publication • https://ieeexplore.ieee.org/abstract/document/8991064

GOVERNMENT PLATFORMS & STANDARDS WE BUILD ON



Ayushman Bharat Digital Mission (ABDM)

National digital health infrastructure — ABHA, Health Information Exchange, FHIR standards.

abdm.gov.in



ICMR Standard Treatment Workflows

Evidence-based clinical protocols behind the Medical Specialist Branch.

icmr.gov.in



Bhashini — National Language Translation Mission

Government voice & translation APIs across 22 Indian languages.

bhashini.gov.in



PM-JAY (Ayushman Bharat)

Central health-cover scheme checked by the eligibility navigator.

pmjay.gov.in



Digital Personal Data Protection Act, 2023

Statutory basis for our consent and data-minimization design.

meity.gov.in



PMBJP — Jan Aushadhi

Generic-medicine pricing behind the cost-savings comparison.

janaushadhi.gov.in